

Week 10

Properties of Convex Functions

Necessary and sufficient conditions for optimality of convex functions

Optimal Convex Functions

Additional Properties

Lagrangian Function

- If first derivative is negative, then decreasing fn. If positive, then increasing.
- If second derivative is negative, then concave; if positive then convex.

Properties of Convex Functions

- If f is a convex function, then all local minima are also global minima.
- Set of all global minima of a convex function is a convex sets.

Necessary and sufficient conditions for optimality of convex functions

- $x^* \in \mathbb{R}^d$ is a global minimum of f iff $\nabla f(x^*) = 0$

Optimal Convex Functions

- Minimize $f(x)$
- f is convex and differentiable

Additional Properties

- If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ are both convex functions, then $h(x) := f(x) + g(x) \forall x$ is also a convex function.
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex and non-decreasing function, $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function and $h(x) = f(g(x))$, then h is a convex function.
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function, $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a linear function and $h(x) = f(g(x))$, then h is a convex function.

Note: In general, if f and g are convex, then $h = f \circ g$ may not be convex.

Lagrangian Function

$$L(x, \lambda) = f(x) + \lambda h(x)$$

, where x is a vector and λ is a scalar.

$$\begin{array}{l} \min_x f(x) \\ \text{s.t. } h(x) \leq 0 \end{array} \equiv \min_x \left[\max_{\lambda \geq 0} L(x, \lambda) \right]$$

- Value at dual optimum $\leq$ Value at primal optimal
- If f, h are convex, then strong duality holds.

Putting it all together

f, h are convex $\Rightarrow$ Strong duality

$\Downarrow$

x^*, λ^* must satisfy. [modulo Regularity conditions]

(a) $\nabla f(x^*) + \lambda^* \nabla h(x^*) = 0$ [Stationarity condition] ✓

(b) $\lambda^* h(x^*) = 0$ [Complementary slackness condition] ✓

(c) $h(x^*) \leq 0$ [Primal feasibility] ✓

(d) $\lambda^* \geq 0$ [Dual feasibility] ✓

KKT Conditions [Karush-Kuhn-Tucker]

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & f_i(x) \leq 0 \quad \forall i=1, \dots, m \\ & g_j(x) = 0 \quad \forall j=1, \dots, n \end{aligned}$$

$$\mathcal{L}(x, u, v) =$$

$$f(x) + \sum_{i=1}^m u_i R_i(x) + \sum_{j=1}^n v_j \mathcal{E}_j(x)$$

(a) $\nabla f(x^*) + \sum_{i=1}^m u_i^* \nabla R_i(x^*) + \sum_{j=1}^n v_j^* \nabla g_j(x^*) = 0 \in \text{Vector}$
 $\rightarrow$ [optimality]

(b) $u_i^* R_i(x^*) = 0 \quad \forall i$ [C.S. condition]

(c) $R_i(x^*) \leq 0 \quad \forall i ; \quad g_j(x^*) = 0 \quad \forall j$ [Primal Feasibility]

(d) $u_i^* \geq 0 \quad \forall i$ [Dual Feasibility]

$$\begin{array}{ll} \text{minimize} & 3x_1 + x_2 \\ \text{subject to} & \begin{cases} x_1 - x_2 + 4 \leq 0 \\ -3x_1 + 2x_2 + 10 \leq 0 \\ x_1, x_2 \geq 0 \end{cases} \end{array}$$

Stationarity conditions $3 + u_1 - 3u_2 = 0$ — **1**

$$1 - u_1 + 2u_2 = 0 \quad \text{--- (2)}$$

Complementary slackness conditions

$$u_1(x_1 - x_2 + 4) = 0 \quad \text{---} \quad (3)$$

$$u_2(-3x_1 + 2x_2 + 10) = 0 \quad (4)$$

Primal feasibility conditions

$$x_1 - x_2 + 4 \leq 0$$

$$-3x_1 + 2x_2 + 10 \leq 0$$

Dual feasibility conditions

$$u_1, u_2 \geq 0$$

From (1) $u_1 = 3u_2 - 3$

Substituting in (2)

$$1 - 3u_2 + 3 + 2u_2 = 0$$

$$\boxed{u_2 = 4}$$

Substituting in

$$u_1 = 9$$

From (3)

$$x_1 - x_2 + 4 = 0$$

$$x_1 = x_2 - 4 \text{ ——— } (5)$$

Substituting in (4)

$$u_2(-3(x_2 - 4) + 2x_2 + 10) = 0$$

$$u_2(-x_2 + 22) = 0$$

$$x_2 = 22 \text{ ——— } (6)$$

Substituting (6) in (5)

$$x_1 = 22 - 4$$

$x_1 = 18$

Minimum value of
objective function is

$$f(x_1, x_2) = 3 \times 18 + 22 = 76$$

Primal/dual solution possibilities

		Primal		
		Finite optimal	Unbounded	Feasible
Dual	Finite optimal	Possible	Impossible	Impossible
	Unbounded	Impossible	Impossible	Possible
	Feasible	Impossible	Possible	Possible

